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PUA528

2. Explain whether each scenario is a classification or regression problem and indicate whether we are most interested in inference or prediction. Finally, provide n and p.

(a) We collect a set of data on the top 500 firms in the US. For each firm we record profit, number of employees, industry, and the CEO salary. We are interested in understanding which factors affect CEO salary.

Answer:

The following factors effect CEO salary:

1. As salary is continuous variable, this problem is Regression problem.
2. We are interested in understanding what is the relation between the salary and the other predictors (such as factors that influence the salary: profit, number of employees, industry)
3. n=500(number of observations in the data set)
4. p=3 (number of predictors: profit, number of employees, industry)

(b) We are considering launching a new product and wish to know whether it will be a success or failure. We collect data on 20 similar products that were previously launched. For each product we have recorded whether it was a success or failure, price charged for the product, marketing budget, competition price, and ten other variables.

Answer:

For the above problem:

1. As the new product is be classified into success or failure, hence it is classification problem.
2. Here the aim is to predict about the new product success or failure, hence it is a prediction of the classification instead of understanding the relationship between predictor and the new product.
3. n=20(number of observations)
4. p=13(number of predictors, price charged for the product, marketing budget, competition price, and ten other variables.)

(c) We are interested in predicting the % change in the USD/Euro exchange rate in relation to the weekly changes in the world stock markets. Hence, we collect weekly data for all of 2012. For
each week we record the % change in the USD/Euro, the % change in the US market, the % change in the British market, and the % change in the German market.

Answer:

From the above problem:

1. Here it is regression problem as the %change in the USD/Euro exchange rate which is the response variable is continuous.
2. Here the aim is to predict %change which is also a continuous numeric value.
3. Here data is collected in weekly for a year (2012) which has 52 weeks hence n=52.
4. Here p=3 which is the predictor variable such as %change in US market, British Market and German Market.

5. What are the advantages and disadvantages of a very flexible (versus a less flexible) approach for regression or classification? Under what circumstances might a more flexible approach be preferred to a less flexible approach? When might a less flexible approach be preferred?

Answer:

For the case of regression model:

The advantage of very flexible approach is it has decreased bias and better fit for non-linear models, also it has lower MSE for test score and thus predict the results accurately.

The disadvantage is less flexible, test score (MSE) is higher, less accurate model.

Circumstances:

When we are trying to predict the results a more flexible approach is preferred, also we need larger dataset in the case of more flexible approach.

For the case of classification model:

The advantage of very flexible approach is training MSE is lower and test MSE is higher, also the variance is very high, alternatively if the model is less flexible it will have lower variance but higher bias.

Circumstances:

A less flexible approach would be preferred when are interested in finding the inference and to interpretate the results with respect to smaller dataset this approach is preferred.

6. Describe the differences between a parametric and a non-parametric statistical learning approach. What are the advantages of a parametric approach to regression or classification (as opposed to a nonparametric approach)? What are its disadvantages?

Answer:

Parametric and non-parametric differences:

Parametric approach reduces the problem of estimating functional form of f to set of parameters. By assuming a parametric form for f simplifies the problem of estimating functional form of f because it is easier to estimate a set of parameters in the linear model than to fit an arbitrary functional form of f.

Non- Parametric approach does not make any assumption about the functional form of f, instead they seek an estimate of f that gests closer to the data point without being too rough or wiggly. This type of approach has potential to accurately fit a wider range of possible shapes for f by avoiding the assumption of a particular functional form for f.

Parametric and non-parametric advantages and disadvantages:

Advantages of a parametric approach to a regression or classification are the simplifying of modelling f to a few parameters and we don't require many observations which is not the case of non-parametric approach.

PARTHA SAI MADALLAPALLI
PUA528

Disadvantages of a parametric approach to a regression or classification is that the model we chose will usually not match the true unknown form of functional f, if the chosen model is far from the functional form of f, then our estimate will be poor with less accuracy.

Advantages of non-parametric approach is to avoid the initial assumption about the form of f is that non-parametric methods can potentially fit more accurate possible shapes f.

For a non-parametric approach, the disadvantage is that we require a much larger number of observations (far more than is typically needed for a parametric approach) for accurate estimate for f, since they do not reduce the problem of estimating f to a small number of parameters.

8. Question:

(a) Use the pd.read_csv() function to read the data into Python. Call the loaded data college. Make sure that you have the directory set to the correct location for the data:

```
[15]: #importing required libraries:
      import numpy as np
      import pandas as pd
      import matplotlib.pyplot as plt
      import seaborn as sns

[2]: #question 8.a
     #import college data:
     college_ds = pd.read_csv('.\college.csv')

[3]: # view top 5 rows:
     college_ds.head()
```

[3]:

	Unnamed: 0	Private	Apps	Accept	Enroll	Top10perc	Top25perc	F.Undergrad	P.Undergrad	Outstate	Room.Board	Books	Personal	PhD	Terminal	S.F.R
0	Abilene Christian University	Yes	1660	1232	721	23	52	2885	537	7440	3300	450	2200	70	78	
1	Adelphi University	Yes	2186	1924	512	16	29	2683	1227	12280	6450	750	1500	29	30	
2	Adrian College	Yes	1428	1097	336	22	50	1036	99	11250	3750	400	1165	53	66	
	Agnes															

PARTHA SAI MADALLAPALLI
PUA528

Assignment_1_PredMod_PUA

Code | Python 3 (ipykernel)

[3]:
```
# view top 5 rows:
college_ds.head()
```

[3]:

	Unnamed: 0	Private	Apps	Accept	Enroll	Top10perc	Top25perc	F.Undergrad	P.Undergrad	Outstate	Room.Board	Books	Personal	PhD	Terminal	S.F.R
0	Abilene Christian University	Yes	1660	1232	721	23	52	2885	537	7440	3300	450	2200	70	78	
1	Adelphi University	Yes	2186	1924	512	16	29	2683	1227	12280	6450	750	1500	29	30	
2	Adrian College	Yes	1428	1097	336	22	50	1036	99	11250	3750	400	1165	53	66	
3	Agnes Scott College	Yes	417	349	137	60	89	510	63	12960	5450	450	875	92	97	
4	Alaska Pacific University	Yes	193	146	55	16	44	249	869	7560	4120	800	1500	76	72	

8. (b) Look at the data used in the notebook by creating and running a new cell with just the code college in it. You should notice that the first column is just the name of each university in a column named something like Unnamed: 0. We don't really want pandas to treat this as data. However, it may be handy to have these names for later. Try the following commands and similarly look at the resulting data frames:

Assignment_1_PredMod_PUA

Code | Python 3 (ipykernel)

[4]:
```
#question 8.b:
college_ds_rm = college_ds.rename({'Unnamed: 0': 'College'},
axis=1)
college_ds_rm = college_ds_rm.set_index('College')
college_ds_rm.head()
```

[4]:

College	Private	Apps	Accept	Enroll	Top10perc	Top25perc	F.Undergrad	P.Undergrad	Outstate	Room.Board	Books	Personal	PhD	Terminal	S.F.Ratio
Abilene Christian University	Yes	1660	1232	721	23	52	2885	537	7440	3300	450	2200	70	78	18.1
Adelphi University	Yes	2186	1924	512	16	29	2683	1227	12280	6450	750	1500	29	30	12.2
Adrian College	Yes	1428	1097	336	22	50	1036	99	11250	3750	400	1165	53	66	12.9
Agnes Scott College	Yes	417	349	137	60	89	510	63	12960	5450	450	875	92	97	7.7
Alaska Pacific University	Yes	193	146	55	16	44	249	869	7560	4120	800	1500	76	72	11.9

8. (c) Use the describe() method of to produce a numerical summary of the variables in the data set.

PARTHA SAI MADALLAPALLI
PUA528

Assignment_1_PredMod_PUA

Code

Python 3 (ipykernel)

[5]:
```
#question 8.c:
college_ds_rm.describe(include=['number'])
```

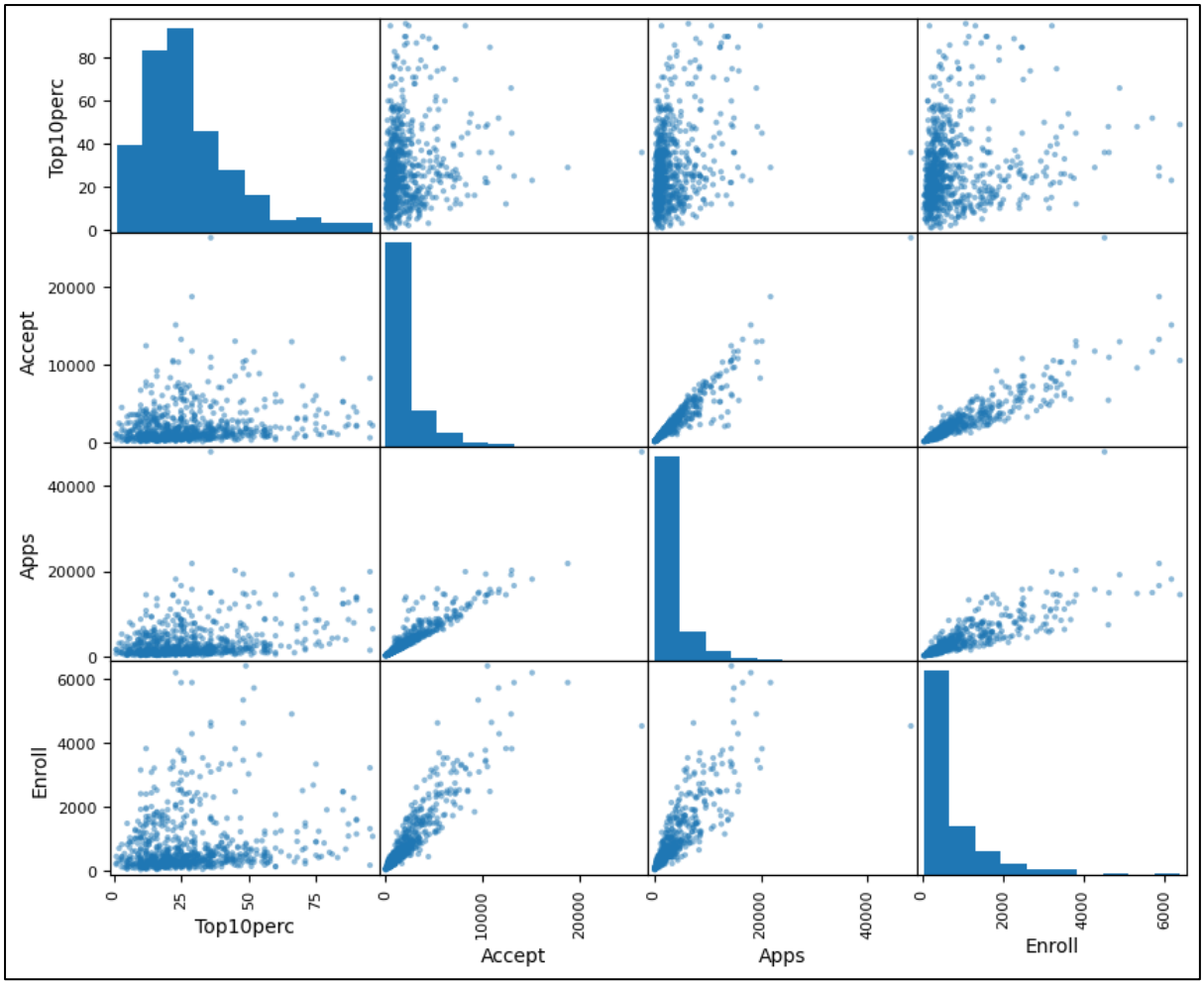
[5]:

	Apps	Accept	Enroll	Top10perc	Top25perc	F.Undergrad	P.Undergrad	Outstate	Room.Board	Books	Personal	
count	777.000000	777.000000	777.000000	777.000000	777.000000	777.000000	777.000000	777.000000	777.000000	777.000000	777.000000	77
mean	3001.638353	2018.804376	779.972973	27.558559	55.796654	3699.907336	855.298584	10440.669241	4357.526384	549.380952	1340.642214	7
std	3870.201484	2451.113971	929.176190	17.640364	19.804778	4850.420531	1522.431887	4023.016484	1096.696416	165.105360	677.071454	1
min	81.000000	72.000000	35.000000	1.000000	9.000000	139.000000	1.000000	2340.000000	1780.000000	96.000000	250.000000	
25%	776.000000	604.000000	242.000000	15.000000	41.000000	992.000000	95.000000	7320.000000	3597.000000	470.000000	850.000000	6
50%	1558.000000	1110.000000	434.000000	23.000000	54.000000	1707.000000	353.000000	9990.000000	4200.000000	500.000000	1200.000000	7
75%	3624.000000	2424.000000	902.000000	35.000000	69.000000	4005.000000	967.000000	12925.000000	5050.000000	600.000000	1700.000000	8
max	48094.000000	26330.000000	6392.000000	96.000000	100.000000	31643.000000	21836.000000	21700.000000	8124.000000	2340.000000	6800.000000	10

8. (d) Use the pd.plotting.scatter_matrix() function to produce a scatterplot matrix of the first columns [Top10perc, Apps, Enroll]. Recall that you can reference a list C of columns of a data frame A using A[C].

[6]:
```
#question 8.d:
columns_of_interest = ['Top10perc', 'Accept','Apps', 'Enroll']
scatter_matrix = pd.plotting.scatter_matrix(college_ds_rm[columns_of_interest], figsize=(10, 8), diagonal='hist')
```

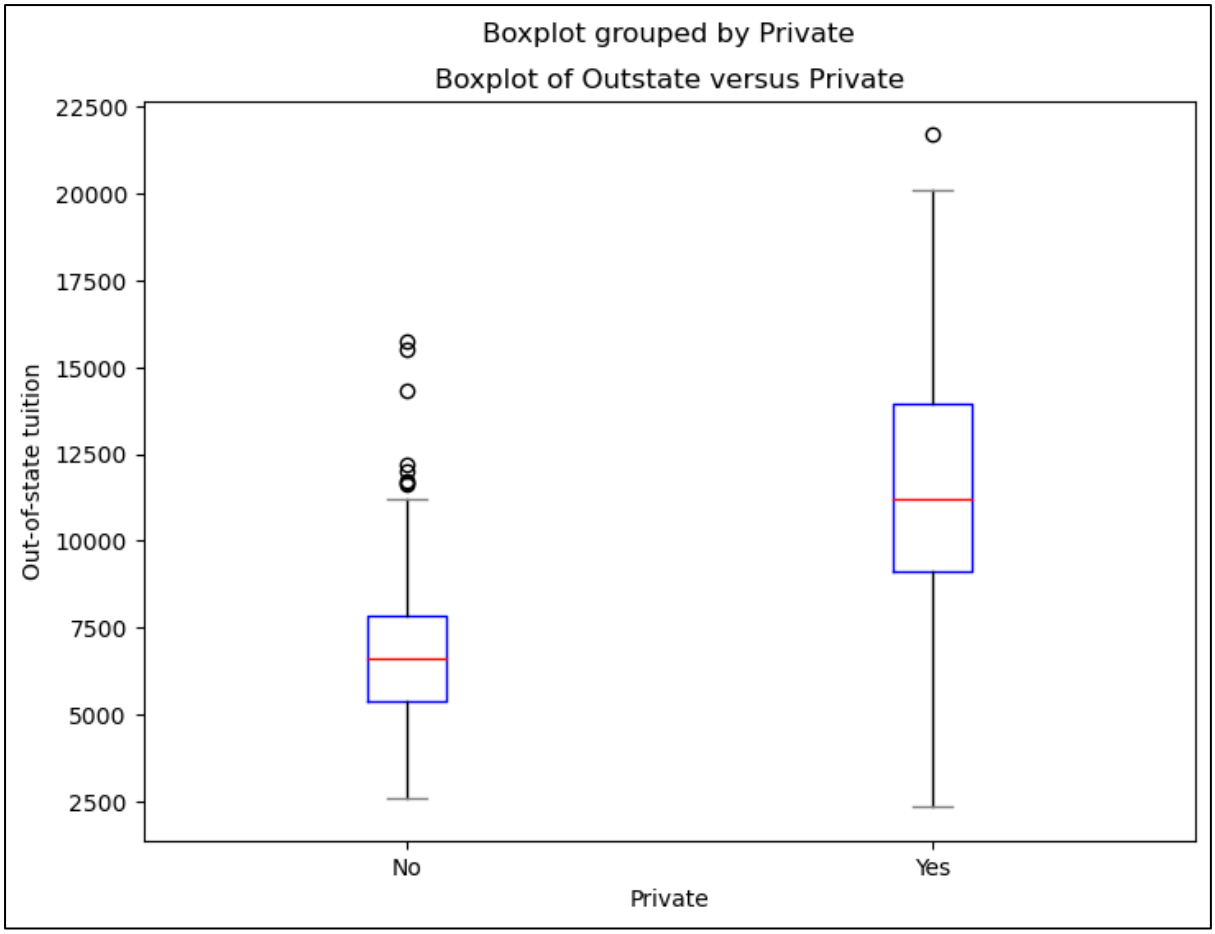
PARTHA SAI MADALLAPALLI
PUA528



8. (e) Use the boxplot() method of college to produce side-by-side boxplots of Outstate versus Private.

```
[7]: #question 8.e:
colors = {'boxes': 'blue', 'whiskers': 'black', 'medians': 'red', 'caps': 'gray'}
college_ds_rm.boxplot(column='Outstate', by='Private', figsize=(8, 6), grid=False,color=colors)
plt.title('Boxplot of Outstate versus Private')
plt.xlabel('Private')
plt.ylabel('Out-of-state tuition')

plt.show()
```

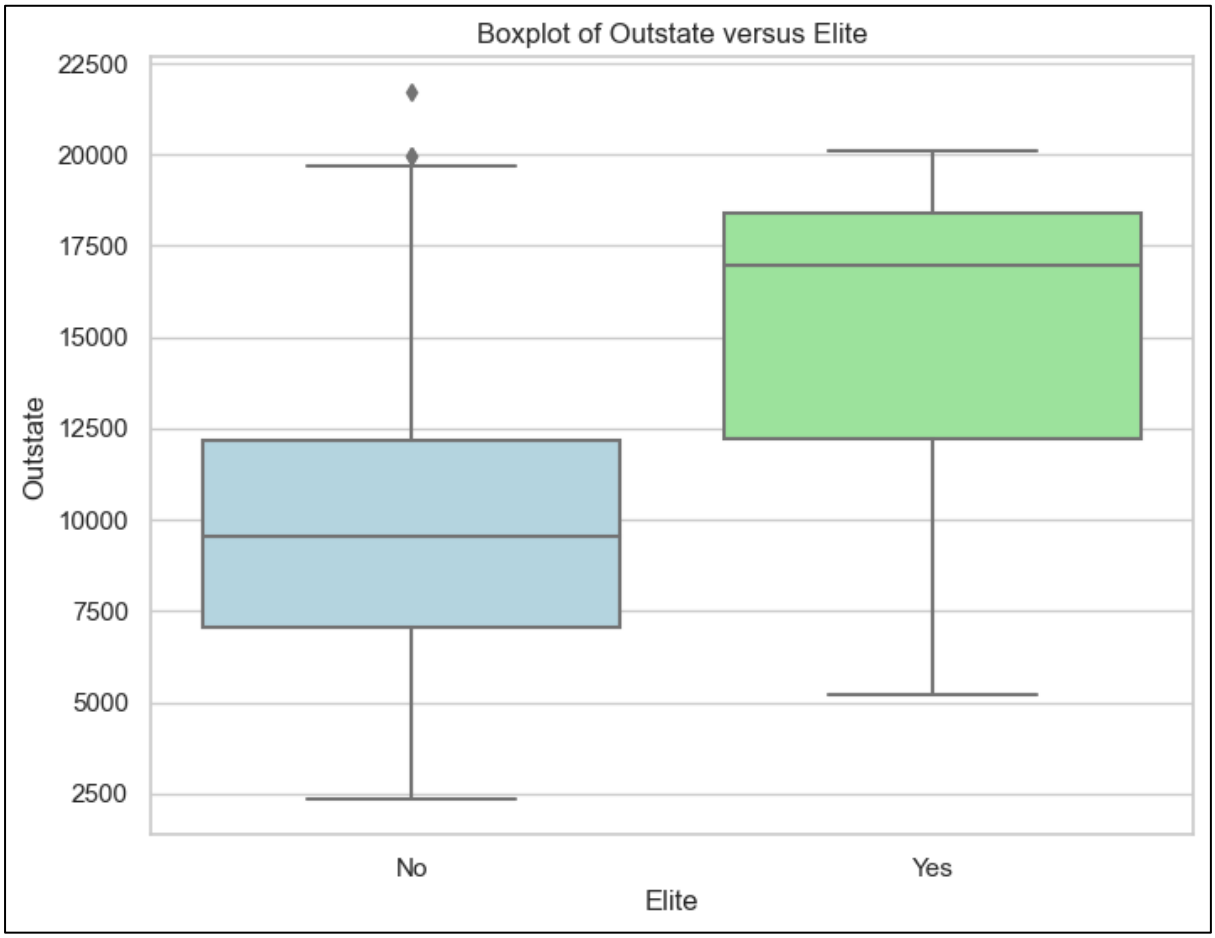
PARTHA SAI MADALLAPALLI
PUA528



8.(f) Create a new qualitative variable, called Elite, by binning the Top10perc variable into two groups based on whether or not the proportion of students coming from the top 10% of their high school classes exceeds 50%.

```
[8]: #Question 8.f:

college_ds['Elite'] = pd.cut(college_ds['Top10perc'],[0,50,100],labels=['No', 'Yes'])

# Check the count of elite universities
elite_counts = college_ds['Elite'].value_counts()
print(elite_counts)

sns.set(style="whitegrid")
plt.figure(figsize=(8, 6))

sns.boxplot(x='Elite', y='Outstate',data=college_ds, palette=['lightblue', 'lightgreen'])

plt.title('Boxplot of Outstate versus Elite')
plt.show()
```

```
Elite
No     699
Yes     78
Name: count, dtype: int64
```

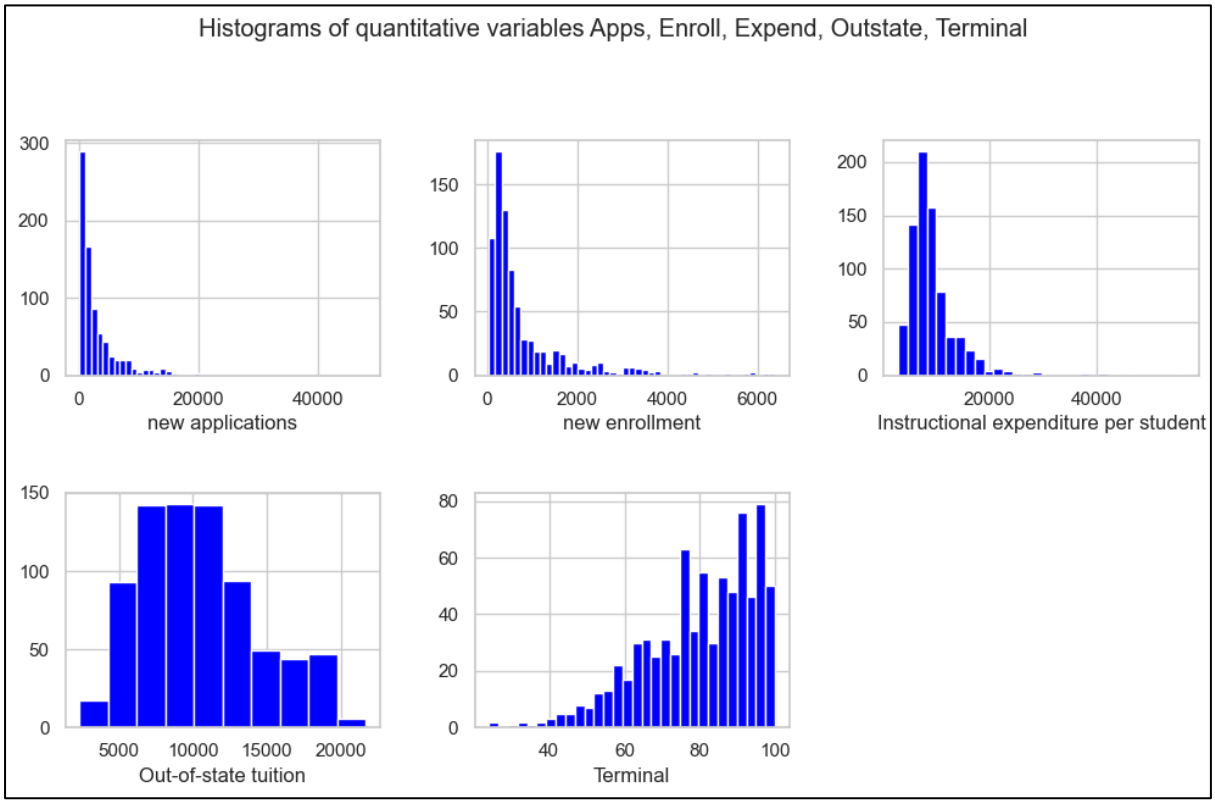


8(g). Use the plot.hist() method of college to produce some histograms with differing numbers of bins for a few of the quantitative variables. The command plt.subplots(2, 2) may be useful:

it will divide the plot window into four regions so that four plots can be made simultaneously. By changing the arguments you can divide the screen up in other combinations.

PARTHA SAI MADALLAPALLI
PUA528

```
[9]: # question 8.g:

fig = plt.figure(figsize=(12, 10))
fig.suptitle('Histograms of quantitative variables Apps, Enroll, Expend, Outstate, Terminal')
fig.add_subplot(3, 3, 1)
plt.hist(college_ds['Apps'], 50, facecolor='blue')
plt.xlabel('new applications')
fig.add_subplot(3, 3, 2)
plt.hist(college_ds['Enroll'], 45, facecolor='blue')
plt.xlabel('new enrollment')
fig.add_subplot(3, 3, 3)
plt.hist(college_ds['Expend'], 30, facecolor='blue')
plt.xlabel('Instructional expenditure per student')
fig.add_subplot(3, 3, 4)
plt.hist(college_ds['Outstate'], facecolor='blue')
plt.xlabel('Out-of-state tuition')
fig.add_subplot(3, 3, 5)
plt.hist(college_ds['Terminal'], 30, facecolor='blue')
plt.xlabel('Terminal')
fig.subplots_adjust(hspace=0.5, wspace=0.3)
plt.show()
```



8. (h) Continue exploring the data, and provide a brief summary of what you discover:

PARTHA SAI MADALLAPALLI
PUA528

```
4]: #question 8.h:

    # Performing Correlation Analysis on PhD and Grad.Rate:

    college_ds = college_ds.dropna(subset=['PhD', 'Grad.Rate'])
    correlation = college_ds[['PhD', 'Grad.Rate']].corr()
    print("Correlation between PhD and Grad.Rate:\n", correlation)

    # Scatter Plot on PhD vs Graduation Rate:

    plt.figure(figsize=(8, 6))
    sns.scatterplot(x='PhD', y='Grad.Rate', data=college_ds)
    plt.title('Scatter Plot of PhD vs Graduation Rate')
    plt.xlabel('Percentage of Faculty with Ph.D.s')
    plt.ylabel('Graduation Rate')
    plt.show()

    # Summary Statistics on PhD vs Grad.Rate:
    summary_stats = college_ds.groupby('PhD')['Grad.Rate'].describe()
    print("Summary Statistics of Graduation Rate by PhD:\n", summary_stats)
```

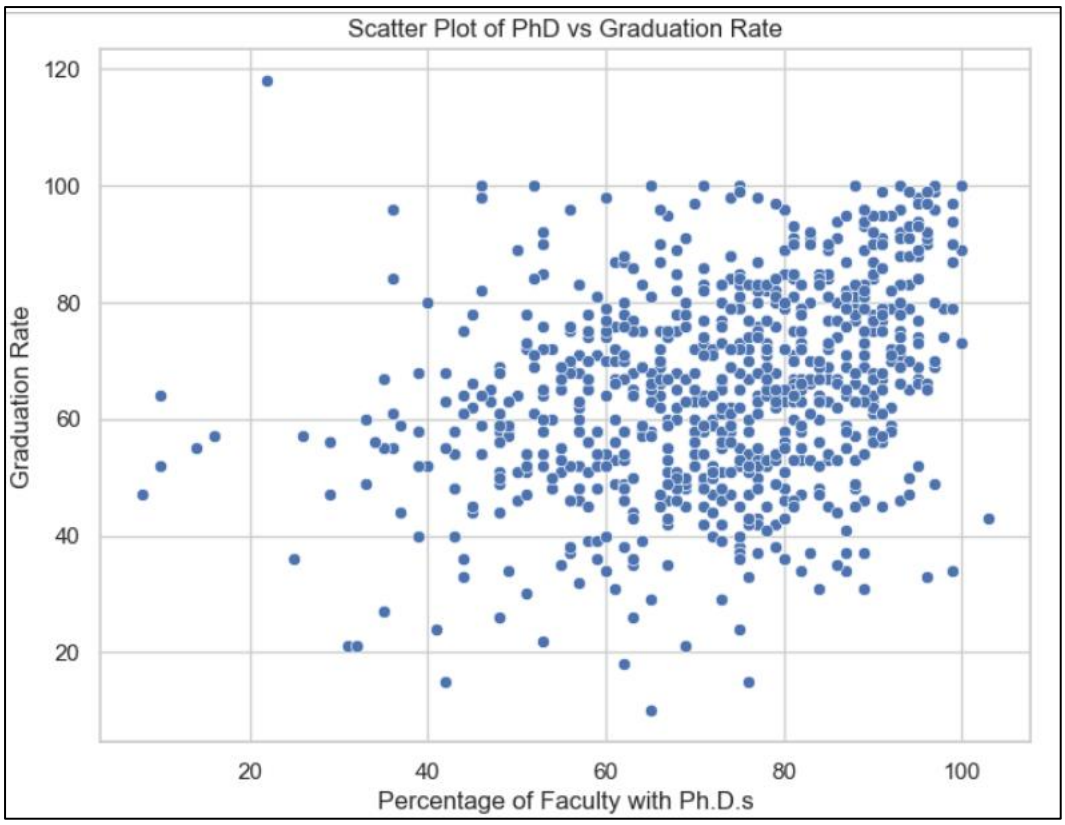
```
Correlation between PhD and Grad.Rate:
                 PhD  Grad.Rate
PhD         1.000000   0.305038
Grad.Rate   0.305038   1.000000
```

Interpretation:

The correlation co-efficient between PhD and Grad.Rate is positive, however it is weak positive linear relationship between this two variables as the correlation co-efficient is 0.305 which is a not a very strong correlation co-efficient.

Thus, universities with a higher percentage of faculty with Ph.D.s tend to have slightly higher graduation rates, but their would be other factors that may also influence graduation rates.

PARTHA SAI MADALLAPALLI
PUA528



Interpretation:

From the above scatter plot the relationship between the percentage of faculty with PhD and the graduation rate for each university from the college dataset. The points in the plot represents an university, with x-axis as faculty with PhD and y-axis with graduation rate.

Here the points are clustered around a diagonal line sloping upwards from left to right which is a positive correlation between faculty with PhD and graduation rate. Also, the spread of dispersion of the points around the diagonal trend line provides insights into the strength of the relationship, as points are tightly clustered around the trend line, it indicates a strong correlation. Also there is one university which has more than 100 faculty with PhD which can be an outlier.

PARTHA SAI MADALLAPALLI
PUA528

```
Summary Statistics of Graduation Rate by PhD:
      count        mean        std    min     25%    50%     75%    max
PhD
8       1.0   47.000000        NaN   47.0   47.00   47.0   47.00   47.0
10      2.0   58.000000   8.485281   52.0   55.00   58.0   61.00   64.0
14      1.0   55.000000        NaN   55.0   55.00   55.0   55.00   55.0
16      1.0   57.000000        NaN   57.0   57.00   57.0   57.00   57.0
22      1.0  118.000000        NaN  118.0  118.00  118.0  118.00  118.0
..      ...         ...        ...    ...     ...    ...     ...    ...
97      7.0   80.428571  19.138592   49.0   69.50   80.0   97.50  100.0
98      2.0   76.500000   3.535534   74.0   75.25   76.5   77.75   79.0
99      6.0   80.166667  23.455632   34.0   81.00   88.5   93.00   97.0
100     3.0   87.333333  13.576941   73.0   81.00   89.0   94.50  100.0
103     1.0   43.000000        NaN   43.0   43.00   43.0   43.00   43.0

[78 rows x 8 columns]
```

Interpretation: From the above statistical summary of Graduation Rate by faculty with PhD qualification across universities. Here the substantial variability in graduation rates, with mean rates ranging from 47% to 118%, followed by there is one observation(university) with faculty a PhD with 8% and its respective university graduation rate is 47%. Also, from the above summary it is understood that the maximum value of 118 for the PhD variable appears to be an outlier and may be considered important for further investigation.

9. This exercise involves the Auto data set studied in the lab. Make sure that the missing values have been removed from the data.

(a) Which of the predictors are quantitative, and which are qualitative?
Answer:

```
[20]: #question 9.a:
      #import Auto data:
      Auto_ds = pd.read_csv('.\Auto.csv')
      Auto_ds_narm = Auto_ds.dropna()

[21]: # view top 5 rows:
      Auto_ds_narm.head()
```

[21]:

	mpg	cylinders	displacement	horsepower	weight	acceleration	year	origin	name
0	18.0	8	307.0	130	3504	12.0	70	1	chevrolet chevelle malibu
1	15.0	8	350.0	165	3693	11.5	70	1	buick skylark 320
2	18.0	8	318.0	150	3436	11.0	70	1	plymouth satellite
3	16.0	8	304.0	150	3433	12.0	70	1	amc rebel sst
4	17.0	8	302.0	140	3449	10.5	70	1	ford torino

PARTHA SAI MADALLAPALLI
PUA528

```
[23]: Auto_ds_narm.dtypes

[23]: mpg             float64
      cylinders         int64
      displacement    float64
      horsepower       object
      weight            int64
      acceleration    float64
      year              int64
      origin            int64
      name             object
      dtype: object

[24]: Auto_ds_narm.horsepower.unique()

[24]: array(['130', '165', '150', '140', '198', '220', '215', '225', '190',
             '170', '160', '95', '97', '85', '88', '46', '87', '90', '113',
             '200', '210', '193', '?', '100', '105', '175', '153', '180', '110',
             '72', '86', '70', '76', '65', '69', '60', '80', '54', '208', '155',
             '112', '92', '145', '137', '158', '167', '94', '107', '230', '49',
             '75', '91', '122', '67', '83', '78', '52', '61', '93', '148',
             '129', '96', '71', '98', '115', '53', '81', '79', '120', '152',
             '102', '108', '68', '58', '149', '89', '63', '48', '66', '139',
             '103', '125', '133', '138', '135', '142', '77', '62', '132', '84',
             '64', '74', '116', '82'], dtype=object)
```

```
[25]: #removing the ? from the data and converting back to numberic:
      Auto_ds_narm = Auto_ds_narm[Auto_ds_narm.horsepower != '?'].copy()
      Auto_ds_narm['horsepower']=pd.to_numeric(Auto_ds_narm['horsepower'])

[26]: Auto_ds_narm.dtypes

[26]: mpg             float64
      cylinders         int64
      displacement    float64
      horsepower        int64
      weight            int64
      acceleration    float64
      year              int64
      origin            int64
      name             object
      dtype: object

[29]: #Quantitative Predictors:

      Quantitative = Auto_ds_narm.select_dtypes(include=['number']).columns
      print(Quantitative)

      Index(['mpg', 'cylinders', 'displacement', 'horsepower', 'weight',
             'acceleration', 'year', 'origin'],
            dtype='object')

      Quantitative predictors are mpg', 'cylinders', 'displacement', 'horsepower', 'weight', 'acceleration', 'year', 'origin'
```

```
[30]: #Qualitative Predictors:

      Qualitative = Auto_ds_narm.select_dtypes(exclude=['number']).columns
      print(Qualitative)

      Index(['name'], dtype='object')

      Qualitative predictors are 'name'
```

(b) What is the range of each quantitative predictor? You can answer this using the min() and max() methods in numpy.

PARTHA SAI MADALLAPALLI
PUA528

```
[32]: # Question 9.b:
      a = Auto_ds_narm.describe()
      a.loc['range'] = a.loc['max'] - a.loc['min']
      a.loc['range']
```

```
[32]: mpg               37.6
      cylinders          5.0
      displacement     387.0
      horsepower       184.0
      weight          3527.0
      acceleration      16.8
      year              12.0
      origin             2.0
      Name: range, dtype: float64
```

(c)What is the mean and standard deviation of each quantitative predictor?

```
[31]: #Question 9.c:
      a = Auto_ds_narm.describe()
      a.loc[['mean','std']]
```

[31]:

	mpg	cylinders	displacement	horsepower	weight	acceleration	year	origin
mean	23.445918	5.471939	194.411990	104.469388	2977.584184	15.541327	75.979592	1.576531
std	7.805007	1.705783	104.644004	38.491160	849.402560	2.758864	3.683737	0.805518

(d) Now remove the 10th through 85th observations. What is the range, mean, and standard deviation of each predictor in the subset of the data that remains?

PARTHA SAI MADALLAPALLI
PUA528

[33]:
```
#question 9.d:

# Removing 10 and 85th records and subsetting the data:
subset_data_rm = Auto_ds_narm.drop(Auto_ds_narm.index[10:85])

# using describe for calculating the mean,range and SD from the subsetted data:
subset_data_rm_des = subset_data_rm.describe()
range_subset = subset_data_rm_des.loc['max'] - subset_data_rm_des.loc['min']
mean_subset = subset_data_rm_des.loc['mean']
std_subset = subset_data_rm_des.loc['std']

# Display the results
print("Range of each predictor in the subset:")
print(range_subset)
print("\nMean of each predictor in the subset:")
print(mean_subset)
print("\nStandard deviation of each predictor in the subset:")
print(std_subset)
```

```
Range of each predictor in the subset:
mpg               35.6
cylinders          5.0
displacement     387.0
horsepower       184.0
weight          3348.0
acceleration      16.3
year              12.0
origin             2.0
dtype: float64

Mean of each predictor in the subset:
mpg               24.374763
cylinders          5.381703
displacement     187.880126
horsepower       101.003155
weight          2938.854890
acceleration      15.704101
year              77.123028
origin             1.599369
Name: mean, dtype: float64
```

```
Standard deviation of each predictor in the subset:
mpg               7.872565
cylinders         1.658135
displacement    100.169973
horsepower       36.003208
weight          811.640668
acceleration      2.719913
year              3.127158
origin            0.819308
Name: std, dtype: float64
```

There isn't much difference in range, standard deviation and mean, after removing the 10th and 85th observations to the original dataset.

(e) Using the full data set, investigate the predictors graphically, using scatterplots or other tools of your choice. Create some plots highlighting the relationships among the predictors. Comment on your findings.

```
: #questioin 9.e:
sns.pairplot(Auto_ds_narm)
```

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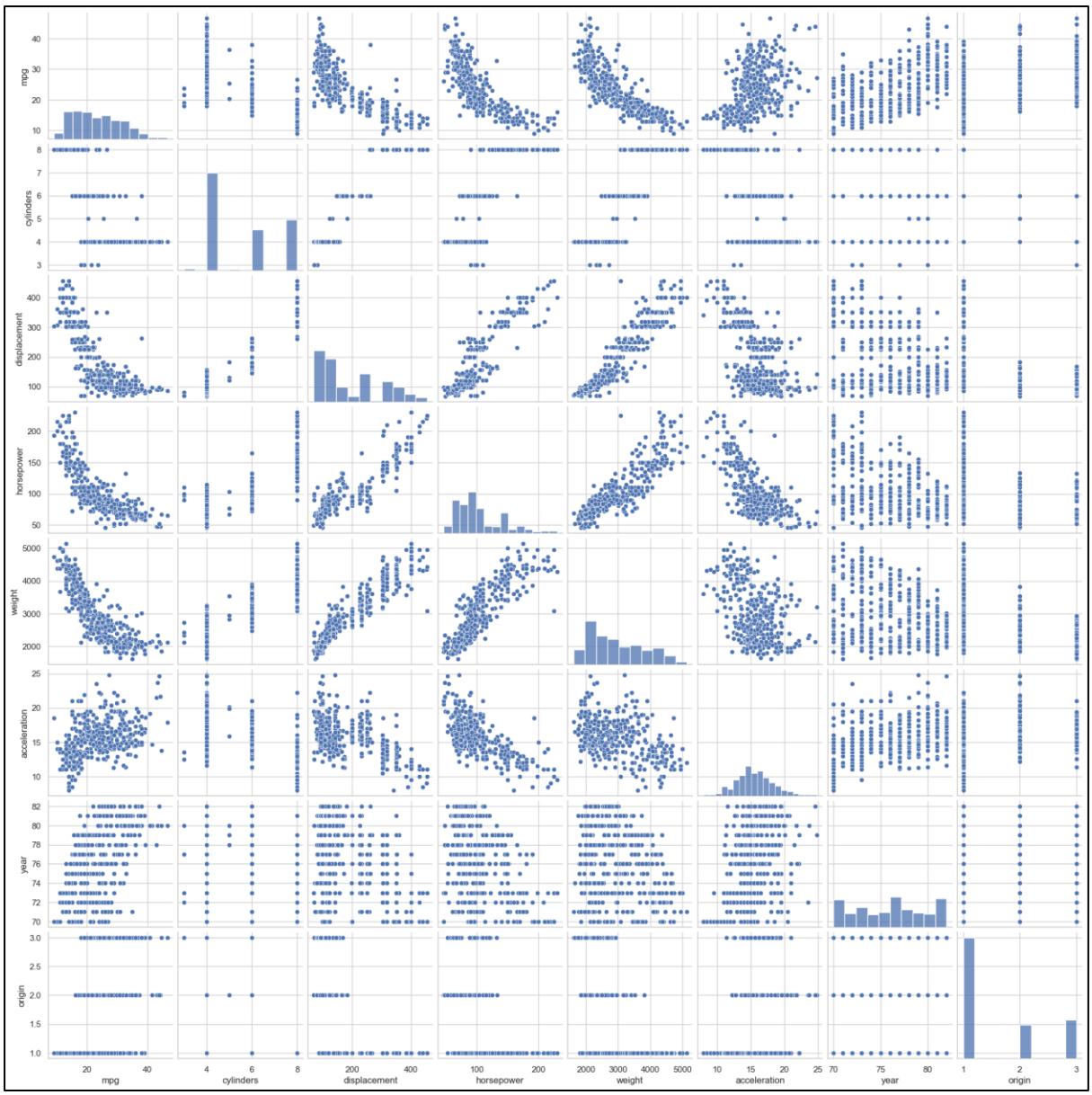


Interpretation:

The variable acceleration follows normal distribution, while displacement and weight have a strong linear relationship. Lastly, mpg has non-linear relationship with weight, horsepower and displacement.

f. Suppose that we wish to predict gas mileage (mpg) on the basis of the other variables. Do your plots suggest that any of the other variables might be useful in predicting mpg? Justify your

answer.

Answer: Yes, with the help of previous question it is understood that displacement, weight and horsepower can play a significant role in prediction of mpg. As displacement is highly correlated with weight we can use this for prediction of mpg.

10. This exercise involves the Boston housing data set.

(a) To begin, load in the Boston data set, which is part of the ISLP library.

```
[69]: #question 10.a:
      #import Boston data:
      Boston_ds = pd.read_csv('.\Boston.csv')
      Boston_ds.drop(Boston_ds.columns[[0]], axis=1, inplace=True)
      Boston_ds.head()
```

[69]:

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	black	lstat	medv
0	0.00632	18.0	2.31	0	0.538	6.575	65.2	4.0900	1	296	15.3	396.90	4.98	24.0
1	0.02731	0.0	7.07	0	0.469	6.421	78.9	4.9671	2	242	17.8	396.90	9.14	21.6
2	0.02729	0.0	7.07	0	0.469	7.185	61.1	4.9671	2	242	17.8	392.83	4.03	34.7
3	0.03237	0.0	2.18	0	0.458	6.998	45.8	6.0622	3	222	18.7	394.63	2.94	33.4
4	0.06905	0.0	2.18	0	0.458	7.147	54.2	6.0622	3	222	18.7	396.90	5.33	36.2

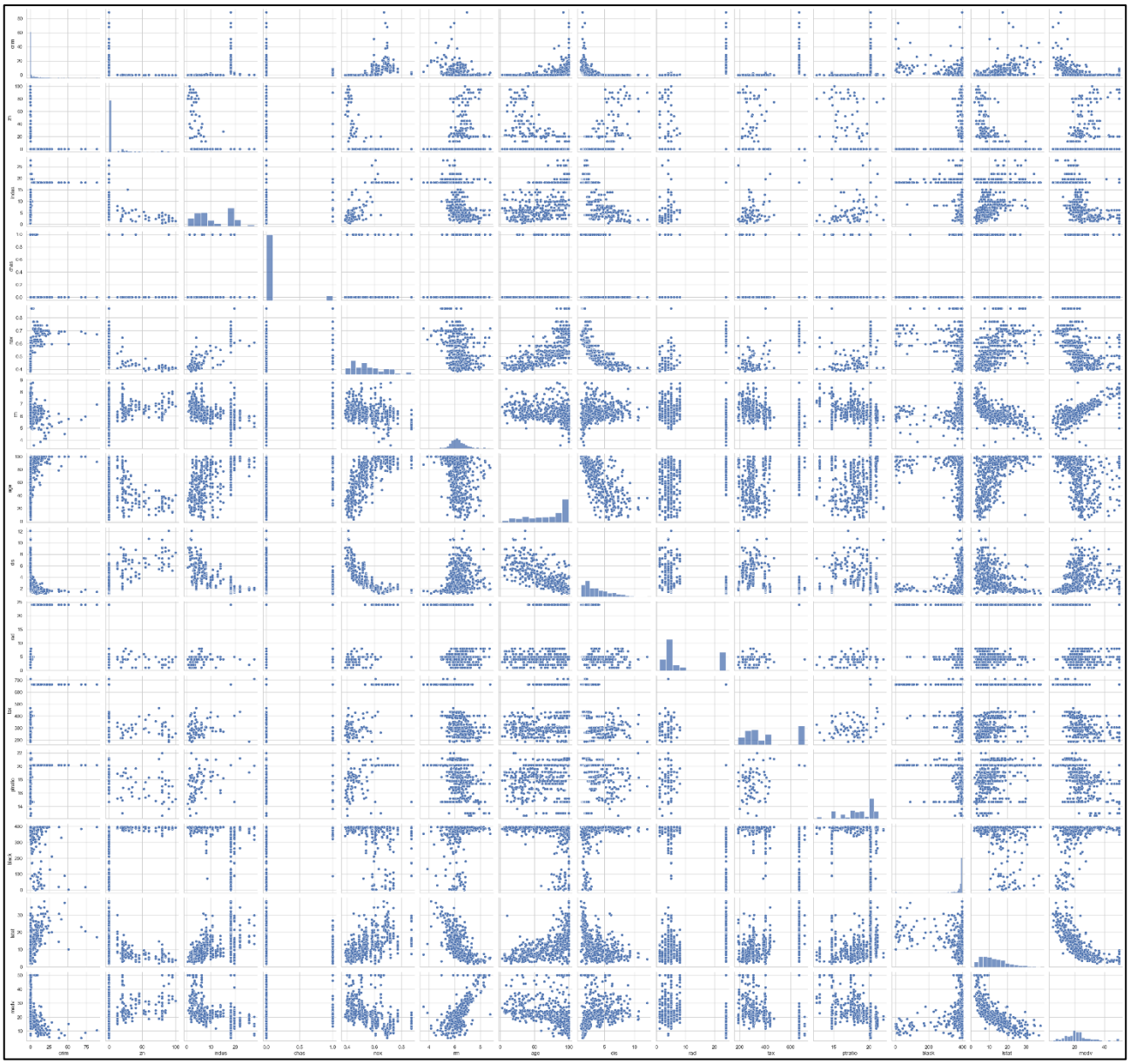
(b)How many rows are in this data set? How many columns? What do the rows and columns represent?

```
0]: #question 10.b:
    print('number of rows',Boston_ds.shape[0])
    print('number of cols',Boston_ds.shape[1])

    number of rows 506
    number of cols 14
```

Each row in the dataset represents a town in the Boston area, while each column represents a feature that may influence house prices. For example, one of the columns, 'CRIM', represents the per capita crime rate by town.

PARTHA SAI MADALLAPALLI
PUA528

(c) Make some pairwise scatterplots of the predictors (columns) in this data set. Describe your findings.

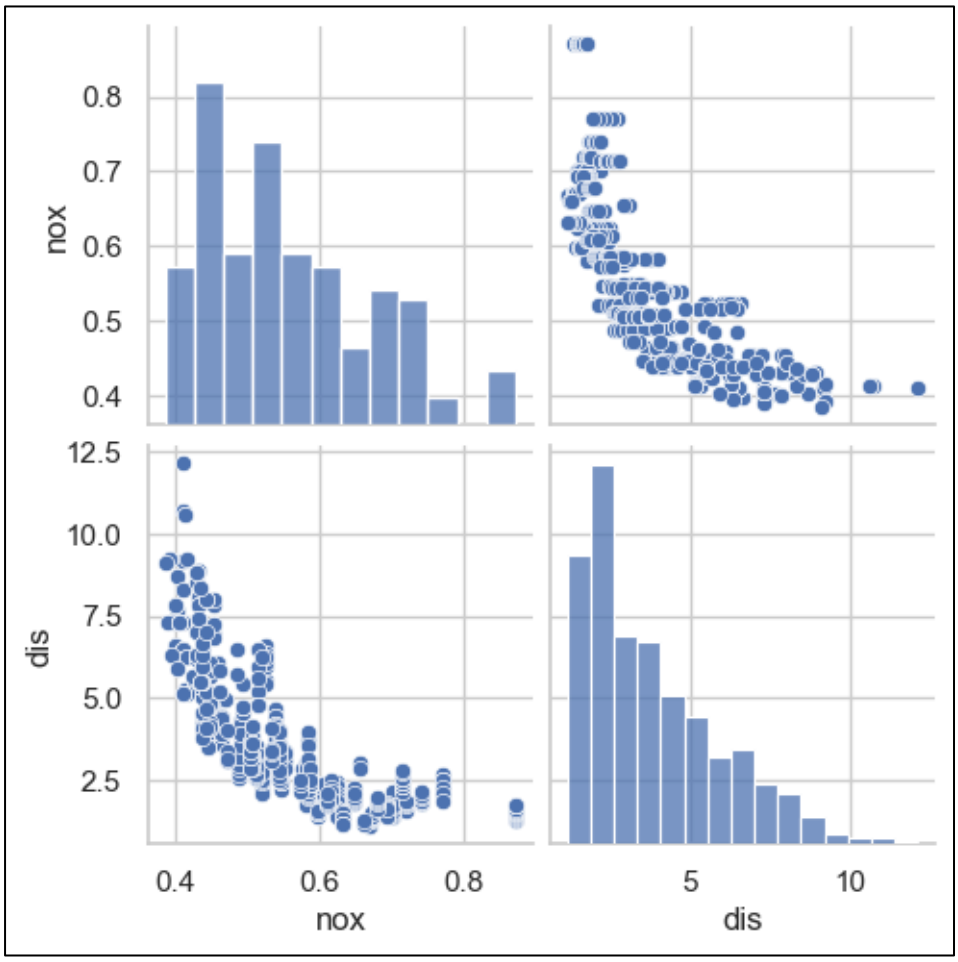


Findings:

A pairwise scatterplot comparing nitric oxides concentration to the weighted distances to five Boston employment centers indicates that as the distance to

the employment centers decreases, the concentration of nitric oxides tends to increase.

```
sns.pairplot(Boston_ds, vars=['nox', 'dis'])
```



(d) Are any of the predictors associated with per capita crime rate? If so, explain the relationship.

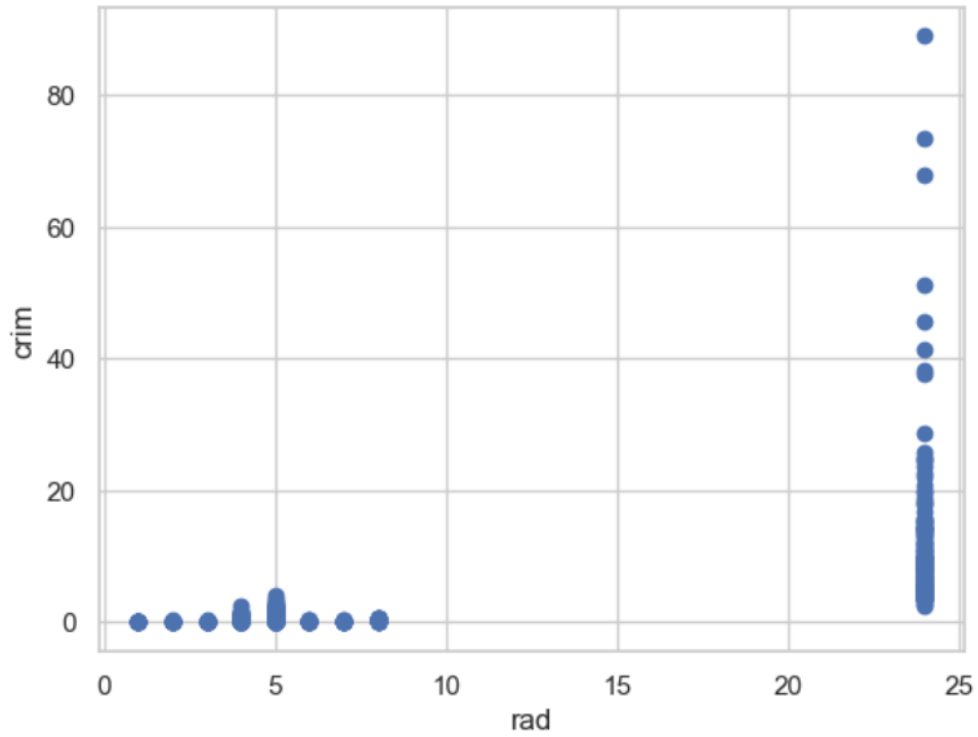
```
#Question 10.d:

Boston_ds.corrwith(Boston_ds['crim']).sort_values()
```

```
medv      -0.388305
black     -0.385064
dis       -0.379670
rm        -0.219247
zn        -0.200469
chas      -0.055892
ptratio    0.289946
age        0.352734
indus      0.406583
nox        0.420972
lstat      0.455621
tax        0.582764
rad        0.625505
crim       1.000000
dtype: float64
```

From the above correlation of each variable with CRIM, we will have a closer largest correlation with RAD,TAX and LSTAT. Hence we will check predictors(RAD,TAX, AND LSTAT) association with capita crime rate:

PARTHA SAI MADALLAPALLI
PUA528

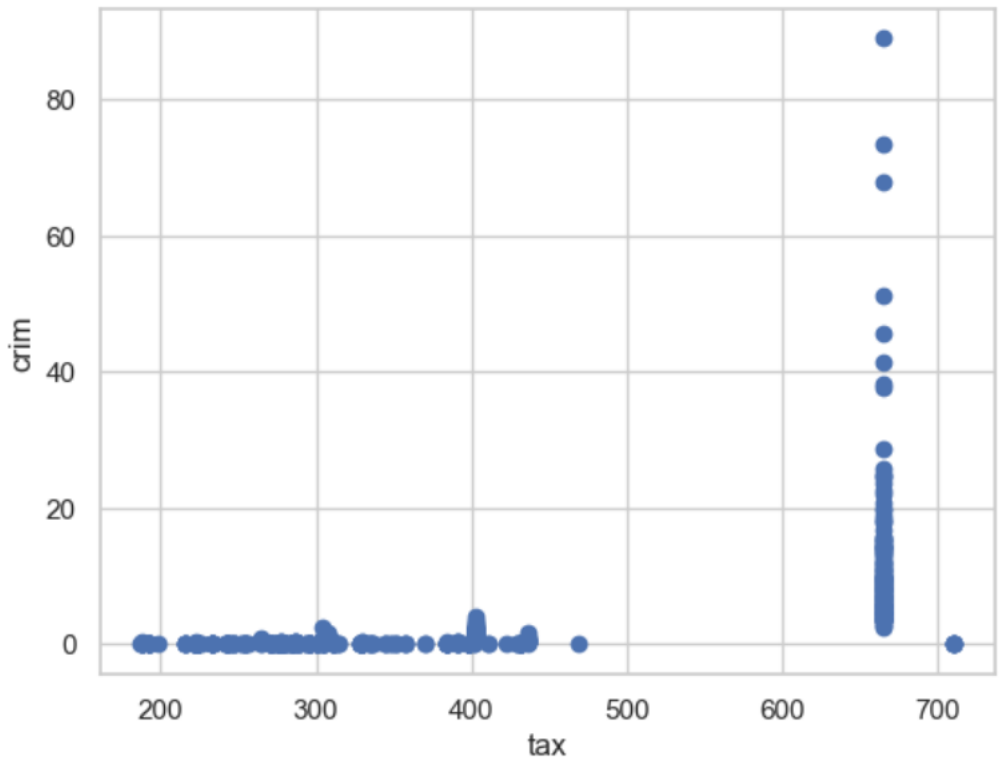
```
[83]: plt.scatter(Boston_ds['rad'], Boston_ds['crim'])
      plt.xlabel('rad')
      plt.ylabel('crim');
```



Findings:
When RAD is around 23 as the highest value, the average CRIM is very higher with large rage.

PARTHA SAI MADALLAPALLI
PUA528

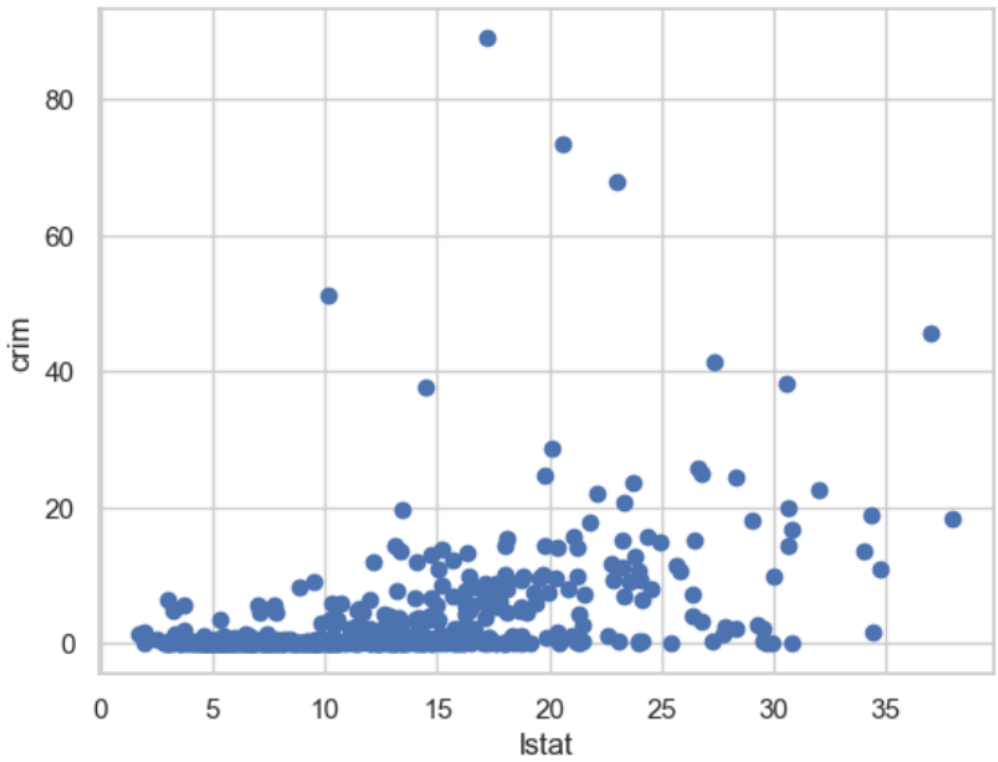
```
plt.scatter(Boston_ds['tax'], Boston_ds['crim'])
plt.xlabel('tax')
plt.ylabel('crim');
```



Findings:

When tax is equal to 600 or above, the average CRIM is much higher and CRIM is much larger.

```
[82]: plt.scatter(Boston_ds['lstat'], Boston_ds['crim'])
      plt.xlabel('lstat')
      plt.ylabel('crim');
```



Findings:

As the LSTAT is around 10, the CRIM is under 10. As the data points spread wider of CRIM. LSTAT which is less than 20 the major proportion of the data points is very close to CRIM with 0.

(e) Do any of the suburbs of Boston appear to have particularly high crime rates? Tax rates? Pupil-teacher ratios? Comment on the range of each predictor.

```
[85]: # question 10.e:
      print("crime rate in suburbs of Boston: " + str(Boston_ds[Boston_ds['crim'] > 20].shape[0]))
      print(" tax rate  rate in suburbs of Boston: " + str(Boston_ds[Boston_ds['tax'] > 600].shape[0]))
      print("pupil-teacher ratio  rate in suburbs of Boston: " + str(Boston_ds[Boston_ds['ptratio'] > 20].shape[0]))

      crime rate in suburbs of Boston: 18
       tax rate  rate in suburbs of Boston: 137
      pupil-teacher ratio  rate in suburbs of Boston: 201
```

(f) How many of the suburbs in this data set bound the Charles

river?

```
# question 10.f:
print("suburbs in this data set bound the Charles river: " + str(Boston_ds[Boston_ds['chas'] == 1].shape[0]))
```

```
suburbs in this data set bound the Charles river: 35
```

(g) What is the median pupil-teacher ratio among the towns in this data set?

```
[89]: # question 10.g:

print("median pupil-teacher ratio: " + str(Boston_ds['ptratio'].median()))
```

```
median pupil-teacher ratio: 19.05
```

(h) Which suburb of Boston has lowest median value of owneroccupied homes? What are the values of the other predictors for that suburb, and how do those values compare to the overall ranges for those predictors? Comment on your findings.

```
# question 10.h:
Boston_ds[Boston_ds['medv'] == Boston_ds['medv'].min()].head(1)
```

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	black	lstat	medv
398	38.3518	0.0	18.1	0	0.693	5.453	100.0	1.4896	24	666	20.2	396.9	30.59	5.0

```
[100]: smmry=Boston_ds.describe()
smmry.loc['range'] = smmry.loc['max'] - smmry.loc['min']
smmry
```

[100]:

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	black	lstat	medv
count	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000
mean	3.613524	11.363636	11.136779	0.069170	0.554695	6.284634	68.574901	3.795043	9.549407	408.237154	18.455534	356.674032	12.653063	22.532806
std	8.601545	23.322453	6.860353	0.253994	0.115878	0.702617	28.148861	2.105710	8.707259	168.537116	2.164946	91.294864	7.141062	9.197104
min	0.006320	0.000000	0.460000	0.000000	0.385000	3.561000	2.900000	1.129600	1.000000	187.000000	12.600000	0.320000	1.730000	5.000000
25%	0.082045	0.000000	5.190000	0.000000	0.449000	5.885500	45.025000	2.100175	4.000000	279.000000	17.400000	375.377500	6.950000	17.025000
50%	0.256510	0.000000	9.690000	0.000000	0.538000	6.208500	77.500000	3.207450	5.000000	330.000000	19.050000	391.440000	11.360000	21.200000
75%	3.677083	12.500000	18.100000	0.000000	0.624000	6.623500	94.075000	5.188425	24.000000	666.000000	20.200000	396.225000	16.955000	25.000000
max	88.976200	100.000000	27.740000	1.000000	0.871000	8.780000	100.000000	12.126500	24.000000	711.000000	22.000000	396.900000	37.970000	50.000000
range	88.969880	100.000000	27.280000	1.000000	0.486000	5.219000	97.100000	10.996900	23.000000	524.000000	9.400000	396.580000	36.240000	45.000000

Findings:

Here the ID 398 suburb of Boston has lowest median value of owneroccupied homes; Compared to other towns, this suburb exhibits high crime rates (CRIM), above-average (INDUS), and elevated levels of nitrogen oxides (NOX). It also features maximum values for the proportion of (AGE), accessibility to (RAD), and the proportion of Black. Additionally, it has a lower-than-average

proportion of (ZN), below-average room numbers per (RM), and above-average levels of (LSTAT).

(i) In this data set, how many of the suburbs average more than seven rooms per dwelling? More than eight rooms per dwelling? Comment on the suburbs that average more than eight rooms per dwelling.

```
[102]: #question 10.i:

print("suburbs average more than seven rooms per dwelling: " + str(Boston_ds[Boston_ds['rm'] > 7].shape[0]))
print("suburbs average more than eight rooms per dwelling: " + str(Boston_ds[Boston_ds['rm'] > 8].shape[0]))

suburbs average more than seven rooms per dwelling: 64
suburbs average more than eight rooms per dwelling: 13

[103]: #suburbs that average more than eight rooms per dwelling:
Boston_ds[Boston_ds['rm']>8].describe()
```

[103]:

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	black	lstat	medv
count	13.000000	13.000000	13.000000	13.000000	13.000000	13.000000	13.000000	13.000000	13.000000	13.000000	13.000000	13.000000	13.000000	13.000000
mean	0.718795	13.615385	7.078462	0.153846	0.539238	8.348538	71.538462	3.430192	7.461538	325.076923	16.361538	385.210769	4.310000	44.200000
std	0.901640	26.298094	5.392767	0.375534	0.092352	0.251261	24.608723	1.883955	5.332532	110.971063	2.410580	10.529359	1.373566	8.092383
min	0.020090	0.000000	2.680000	0.000000	0.416100	8.034000	8.400000	1.801000	2.000000	224.000000	13.000000	354.550000	2.470000	21.900000
25%	0.331470	0.000000	3.970000	0.000000	0.504000	8.247000	70.400000	2.288500	5.000000	264.000000	14.700000	384.540000	3.320000	41.700000
50%	0.520140	0.000000	6.200000	0.000000	0.507000	8.297000	78.300000	2.894400	7.000000	307.000000	17.400000	386.860000	4.140000	48.300000
75%	0.578340	20.000000	6.200000	0.000000	0.605000	8.398000	86.500000	3.651900	8.000000	307.000000	17.400000	389.700000	5.120000	50.000000
max	3.474280	95.000000	19.580000	1.000000	0.718000	8.780000	93.900000	8.906700	24.000000	666.000000	20.200000	396.900000	7.440000	50.000000

```
Boston_ds.describe()
```

	crim	zn	indus	chas	nox	rm	age	dis	rad	tax	ptratio	black	lstat	medv
count	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000	506.000000
mean	3.613524	11.363636	11.136779	0.069170	0.554695	6.284634	68.574901	3.795043	9.549407	408.237154	18.455534	356.674032	12.653063	22.532806
std	8.601545	23.322453	6.860353	0.253994	0.115878	0.702617	28.148861	2.105710	8.707259	168.537116	2.164946	91.294864	7.141062	9.197104
min	0.006320	0.000000	0.460000	0.000000	0.385000	3.561000	2.900000	1.129600	1.000000	187.000000	12.600000	0.320000	1.730000	5.000000
25%	0.082045	0.000000	5.190000	0.000000	0.449000	5.885500	45.025000	2.100175	4.000000	279.000000	17.400000	375.377500	6.950000	17.025000
50%	0.256510	0.000000	9.690000	0.000000	0.538000	6.208500	77.500000	3.207450	5.000000	330.000000	19.050000	391.440000	11.360000	21.200000
75%	3.677083	12.500000	18.100000	0.000000	0.624000	6.623500	94.075000	5.188425	24.000000	666.000000	20.200000	396.225000	16.955000	25.000000
max	88.976200	100.000000	27.740000	1.000000	0.871000	8.780000	100.000000	12.126500	24.000000	711.000000	22.000000	396.900000	37.970000	50.000000

Findings:

Suburbs with average more than eight rooms per dwelling has very less CRIM rate. INDUS and LSTAT is also lower for suburbs with average more than eight rooms per dwelling.